BASIS VANISHING PROPAGATES TO ALL OPENS AND MAKES HOM PRESHEAVES INTO SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Higher direct images on a basis. Cohomology of Sheaves, Lemma cohomology-lemma-unbounded-describe-higher-direct-images. H^i(Rf_*K) as sheafification of V |-> H^i(f^{-1}(V),K).
- Sections of R SheafHom. Cohomology of Sheaves, Lemma cohomology-lemma-section-RHom-over-U. H^0(f^{-1}V, R SheafHom(K,L)) = Hom_D(K|_{f^{-1}V}, L|_{f^{-1}V}).

Source reference. Original lemma 0D5X: sources/stacks-chow/tags.txt, source lines 16502–16502. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08DK: sources/stacks-chow/tags.txt, source lines 10640–10640. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. Let $\mathcal{B}$ be a basis for the topology on Y.*

(1) *Assume K is in $D(\mathcal{O}_X)$ such that for $V \in \mathcal{B}$ we have $H^i(f^{-1}(V), K) = 0$ for $i < 0$. Then Rf_*K has vanishing cohomology sheaves in negative degrees, $H^i(f^{-1}(V), K) = 0$ for $i < 0$ for all opens $V \subset Y$, and the rule $V \mapsto H^0(f^{-1}V, K)$ is a sheaf on Y.*

(2) *Assume K, L are in $D(\mathcal{O}_X)$ such that for $V \in \mathcal{B}$ we have $\mathrm{Ext}^i(K|_{f^{-1}V}, L|_{f^{-1}V}) = 0$ for $i < 0$. Then $\mathrm{Ext}^i(K|_{f^{-1}V}, L|_{f^{-1}V}) = 0$ for $i < 0$ for all opens $V \subset Y$ and the rule $V \mapsto \mathrm{Hom}(K|_{f^{-1}V}, L|_{f^{-1}V})$ is a sheaf on Y.*

Proof. Lemma 0D5X tells us $H^i(Rf_*K)$ is the sheaf associated to the presheaf $V \mapsto H^i(f^{-1}(V), K) = H^i(V, Rf_*K)$. The assumptions in (1) imply that Rf_*K has vanishing cohomology sheaves in degrees < 0. We conclude that for any open $V \subset Y$ the cohomology group $H^i(V, Rf_*K)$ is zero for $i < 0$ and is equal to $H^0(V, H^0(Rf_*K))$ for $i = 0$. This proves (1).

To prove (2) apply (1) to the complex $R\mathcal{H}om(K, L)$ using Lemma 08DK to do the translation. □

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